

SECTION C

11 (**long**) The step response of a linear system is given by

$$\begin{cases} 0 & \text{for } t < 0 \\ A + Be^{-t} + Ce^{-2t} & \text{for } t \geq 0 \end{cases}$$

with constants A , B , and C .

(a) Explain why the system must be second order. Hence, show that its impulse response is given by

$$g(t) = \begin{cases} 0 & \text{for } t < 0 \\ D(e^{-t} - e^{-2t}) & \text{for } t \geq 0 \end{cases}$$

where $D = 2C$.

[10]

(b) Find the system response $y(t)$ to the input

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \\ e^{-t} & \text{for } t \geq 0 \end{cases}$$

for all t .

[10]

(c) For $T > 0$, find the system response $y(t)$ to the input

$$f(t) = \begin{cases} e^{-t} & \text{for } T \leq t \leq 2T \\ 0 & \text{otherwise} \end{cases}$$

for all t .

[10]